```
next_x, acq_val = botorch.optim.optimize.optimize_acqf(
    policy,
    bounds=torch.tensor([[-bound], [bound]], dtype=torch.float),
    q=1,
    num_restarts=10,
    raw_samples=20,
)
```

The returned values are the location of the point giving the highest acquisition score that L-BFGS finds and the corresponding maximized acquisition score. Upon inspection, we have:

```
>>> next_x, acq_val

(tensor([[0.5985]]), tensor(0.9129))
```

This output indicates that the candidate point maximizing the PoI score at 0.91 probability of improvement is at roughly 0.6, which corresponds to the dotted vertical line in Figure 4.12. This point is where we will query the objective function (that is, evaluate the function) to collect the next data point in BayesOpt.

And that is all there is to deciding which point to query using the PoI policy at a single iteration of BayesOpt. But remember the BayesOpt loop in Figure 4.2, where we alternate between finding the next data point to query using a policy (Step 2), and updating our GP with that new data (Steps 1 and 3). We will do this in section 6.2.3.

4.2.3 Diagnosing the Probability of Improvement policy

In this subsection, we finally run the PoI policy and analyze its behavior. Again, we will repeat the entire process—training the model, declaring the PoI policy, and using `optimize_acqf()` to find the best point—multiple times until we reach a termination condition. As we saw, this loop is implemented in notebook `01 - BayesOpt loop.ipynb`. What we need to do now is initalize the PoI policy within the appropriate for loop.

This code will produce a series of plots generated by the helper function `visualize_gp_belief_and_policy()`, each showing the current state of our BayesOpt loop throughout the ten queries that we make. These plots look similar to Figure 4.12, with the addition of the objective function for our reference.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>